Introduction to Economics

Class 8 — Surplus, efficiency, taxes and subsidies

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Plan for today

Warm-up; Class 6's coffee: just right?
Bread, milk, eggs: VAT down to zero
What buyers gain: consumer surplus
What sellers gain: producer surplus
In pairs: menús del día
Total surplus: efficient, but fair?
A tax: a wedge between two prices
Break
Who pays the tax office; where the money goes
In pairs: a tax on menús
Who bears a tax: elasticity decides; in pairs
A subsidy; VAT; the bread answered
In one line; before Thursday 22 October

Slide 1 Title

Preview version. The full slides and handout are posted after class.

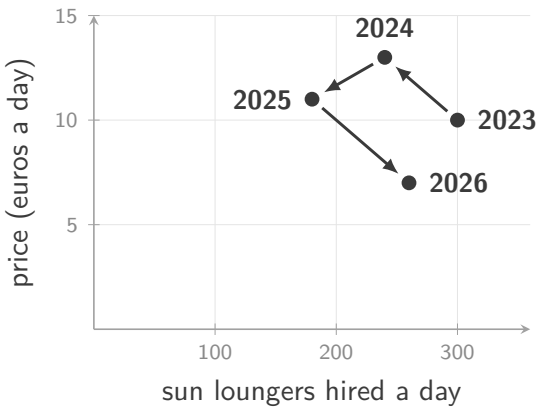
Block 3 begins. Blocks 1 and 2 asked how a market works: where it settles, which way it moves and how far. Block 3 asks whether the outcome is any good, and what a government does to it. The first half builds a way to measure what a market is worth to its buyers and its sellers, in euros, and shows why the market's own quantity makes that total as large as it can be. The second half puts a tax on the same market: who ends up paying it, where the money goes, and what a subsidy does; the thread through the class is a real VAT cut on bread, milk and eggs.

Slide 2 Plan for today

The class at a glance. The first half answers one question about Class 6's market: is its quantity any good? The tool is surplus, a way to put a buyer's gain and a seller's gain in euros; adding the two gives what the market is worth, and the market's own quantity turns out to make that as large as possible. The second half puts a tax on the same market: who bears it, where the money goes, and a subsidy as the same thing in reverse; a real VAT cut on basic food opens the class as a question and closes it as the answer.

Two halves of fifty minutes with a ten-minute break between them; two short warm-ups on earlier classes come first.

Warm-up: two dots — which curve moved?



Sun loungers on one beach, each August

Each year only one curve moved. Which one, which way?

1. 2023 → 2024
2. 2024 → 2025
3. 2025 → 2026

In pairs, two minutes; then hands per arrow.

warm-up 1, not numbered

Warm-up 1 Two dots: which curve moved?

A warm-up on reading a market backwards. Each dot is one August's equilibrium in the market for sun loungers on one beach, and between two years only one curve moved. Before class, decide for each arrow whether the price rose or fell and whether more or fewer loungers were hired, then which curve must have moved: the new dot lies on the curve that stayed put.

Revealed in class.

Warm-up: what does playing cost you?

Saturday: a volleyball tournament

Entry, already paid: 12 euros, no refund
Same hours: an usher's shift at a film festival, 40 euros
Bus there and back: 4 euros

1. What does playing cost you?
2. Playing is worth 42 to you. Play?

In pairs, one minute; then hands.

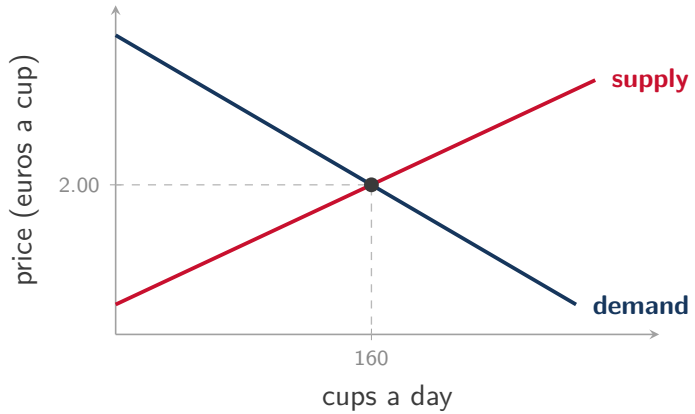
warm-up 2, not numbered

Warm-up 2 What does playing cost you?

A warm-up on opportunity cost, from Class 2. Before class, list what you give up by playing and what you pay only because you play; then ask whether money already paid, and not refundable, belongs on that list. Item 2 compares that cost with what playing is worth to you.

Revealed in class.

Class 6's coffee: 160 cups — too few, too many, or just right?



Slide 3 Class 6's coffee: 160 cups, too few, too many, or just right?

Class 6's market, with its crossing, asked a new question. Class 6 showed that the price finds 160 cups and 2.00 on its own; it never asked whether 160 is a good number of cups. Opinions in the room differ, and that is the point: without a measure, "too few" and "too many" are just tastes. Slides 5 to 9 build the measure, in euros, and slides 10 and 11 come back to this picture with an answer.

1 January 2023: VAT on bread, milk and eggs fell from 4 % to 0 %

4 % → 0 %

bread, milk, cheese, eggs, fruit, vegetables

10 % → 5 %

olive and seed oils, pasta

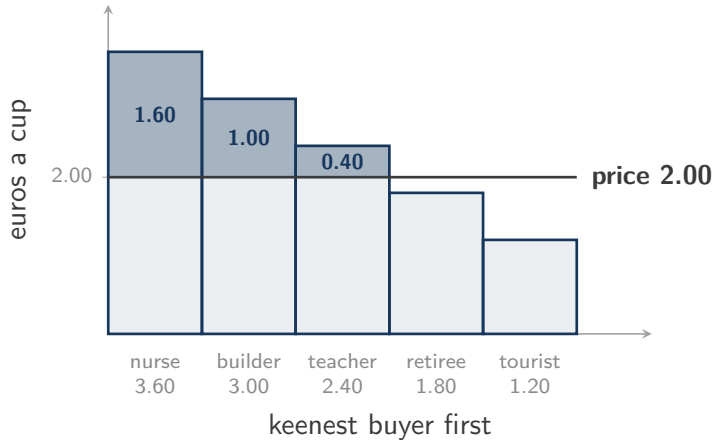
A 1.04 € loaf: 4 cents of VAT.
How many reached the shopper?

Real Decreto-ley 20/2022 (BOE, 28 December 2022), in force 1 January 2023.

Slide 4 1 January 2023: VAT on bread, milk and eggs fell from 4 % to 0 %

The hook, from the official gazette. From the start of 2023 the government took the VAT off the most basic foods and halved it on oils and pasta, to fight food inflation. On a loaf that cost 1.04, four cents were VAT; after the cut the shop handed nothing to the tax office for it. Whether the loaf then cost 1.00, or the shop kept some of the four cents, is the question the second half answers; consumer groups at the time said many shops had not passed it on, and the papers ran the story. Keep the question in mind while the tools are built: it needs a way to measure who gains from a market.

Willingness to pay: each step of demand is one buyer



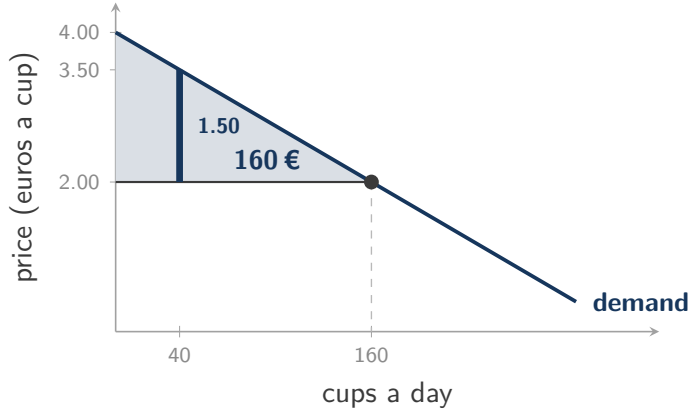
Consumer surplus = willingness to pay − price: 1.60 + 1.00 + 0.40 = 3.00 €.

Slide 5 Willingness to pay: each step of demand is one buyer

Five buyers at one counter, each with the most they would pay this morning; that maximum is the willingness to pay, how much a buyer values the good, in euros. Lined up keenest first, the five steps are a demand curve: at any price, the buyers whose step rises above it buy, the others do not. What a buyer would have paid minus what she actually pays is her gain from the trade, her consumer surplus; the two whose steps lie below the price do not buy and gain nothing.

Try it yourself: drop the price to 1.50 and recount. The surplus grows in two ways, because the old buyers pay less and a new buyer joins.

Consumer surplus: under demand, above the price — 160 € a day



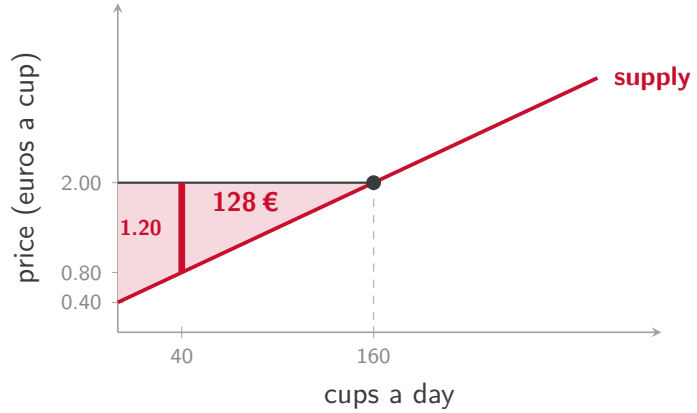
½ × 160 cups × 2.00 € = 160 euros a day: every buyer's bargain, added up.

Slide 6 Consumer surplus: under demand, above the price, 160 € a day

The staircase with 320 steps, each one cup wide, is Class 5's straight line, and its height at any cup is the willingness to pay of whoever buys that cup. The bar at the 40th cup is one buyer's gain, 3.50 minus the price; the buyer of the 160th cup pays exactly what it is worth to him and gains nothing: the marginal buyer, the first to leave if the price rose. Adding every bar gives the triangle, and a triangle's area is half the base times the height; forgetting the half is the usual slip.

The other slip is to take the box below the price, 2.00 times 160 cups, for the surplus. That box is what buyers spend, Class 7's revenue; the surplus is what they would have paid on top of it, above the price.

Producer surplus: above supply, below the price — 128 € a day



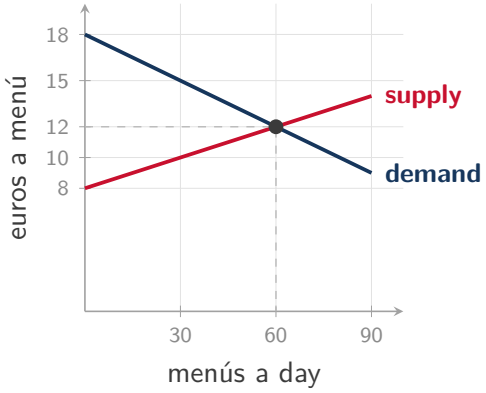
½ × 160 cups × 1.60 € = 128 euros a day: what the bars get above their costs.

Slide 7 Producer surplus: above supply, below the price, 128 € a day

The mirror image for sellers. A bar makes one more cup only if the price covers that cup's cost, everything the bar gives up to make it, the owner's own time included; so the height of supply at any cup is the cost of the bar that makes it, and the bar making the 160th cup is the marginal seller, the first to stop if the price fell. The price received minus the cost, added over all the cups, is the triangle above supply and below the price; its height is the price minus 0.40, the lowest cost on the line.

Producer surplus is close to profit but not the same: the bar's rent is paid whether or not this cup is made. Costs are treated properly in Class 11.

In pairs, four minutes: read the surplus off the graph



Menús del día on one street

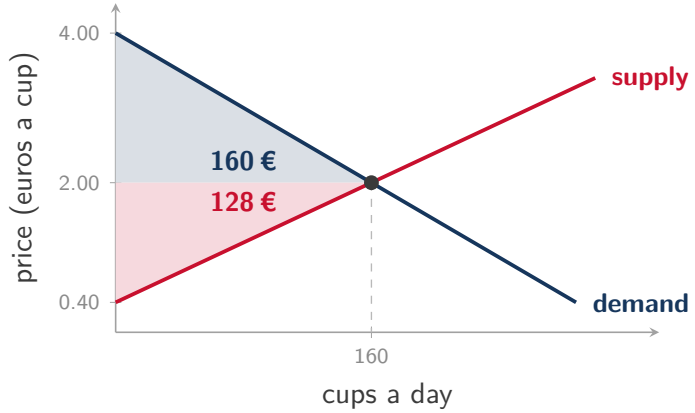
1. Consumer surplus at 12 €?
2. Producer surplus at 12 €?
3. The 30th menú: worth? costs?

Slide 8 In pairs, four minutes: read the surplus off the graph

Exercise 1, on a street of menús del día rather than the coffee market. Before class, read the two surplus triangles off the graph: each starts at the price and ends where its own curve meets the price axis, and both need the half. For item 3, go up from the 30th menú to each curve and ask what its buyer would pay and what the bar that makes it needs. Check that none of your answers is what buyers spend.

Revealed in class.

Total surplus: buyers' 160 plus the bars' 128 — 288 euros a day

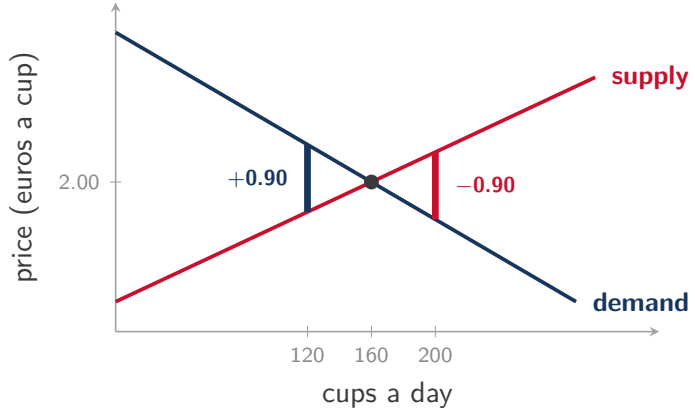


Total surplus = value to buyers − cost to sellers = 160 + 128 = 288 euros a day.

Slide 9 Total surplus: buyers' 160 plus the bars' 128, 288 euros a day

The two triangles together fill the area between the curves up to the market quantity, and that area is what the market is worth: the value of the cups to those who drink them, the whole area under demand, minus what they cost the bars that make them, the area under supply. The price does not appear in the total, because what buyers pay is exactly what the bars receive; a higher price only moves surplus from the blue triangle to the red one. That is why the next question is about the quantity, not the price.

Stop at 160: before it, cups worth more than they cost; after it, less



Below 160: surplus left on the table. Above: cups that cost more than they are worth.

Slide 10 Stop at 160: before it, cups worth more than they cost; after it, less

Why the market quantity is the right one, read one cup at a time. For any cup to the left of the crossing the demand line sits above the supply line: someone values that cup at more than it costs a bar to make, so not making it throws surplus away, 0.90 of it at the 120th cup. For any cup to the right, supply sits above demand: making it uses up more than it is worth. Only at the crossing is the next cup worth exactly its cost, so no other quantity gives a larger total. The 0.90 gap at 120 cups comes back three slides on, as a tax.

Just right: the market's 160 cups give the largest total surplus

- The cups go to those who value them most
- They are made by the bars that make them cheapest
- Every cup worth more than its cost, and no other

That is what economists call **efficient**, as long as no trade affects others (Class 10).

Slide 11 Just right: the market's 160 cups give the largest total surplus

Slide 3's answer. The three lines are the reading's three reasons, and each one is a sorting: at 2.00 the cups go exactly to the people who value one at 2.00 or more, so nobody who values a cup less takes it from someone who values it more; the bars that make them are exactly those that can do so for 2.00 or less; and slide 10 settled the quantity. An outcome that makes total surplus as large as it can be is called efficient, and the market reaches it with nobody planning it.

Two conditions sit behind this: everyone is a price taker, and the trade affects nobody outside it. When a bar's terrace keeps the street awake, that cost is on no curve; that is Class 10.

Efficient is not the same as fair

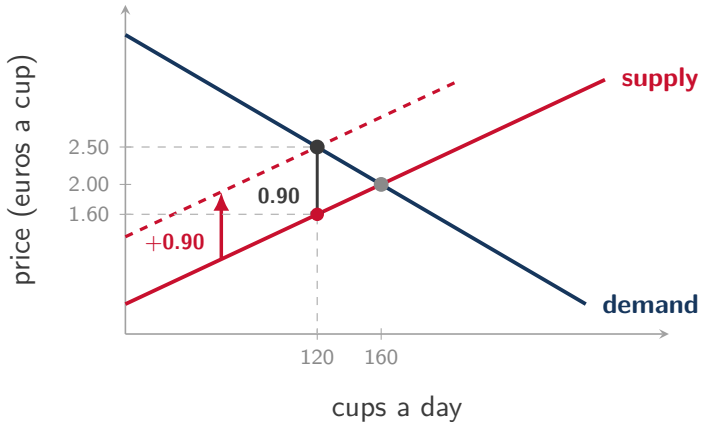
- Efficiency: is the cake as big as it can be?
- Equity: is it shared fairly?
- Willingness to pay depends on income

Efficiency is a positive question; fairness a normative one (Class 4).

Slide 12 Efficient is not the same as fair

Two different questions about the same cake. Whether the cake is as big as it can be is measurable, so efficiency is a positive claim; whether it is cut fairly is a judgement, so equity is normative, Class 4's distinction. The third line says why the two can come apart: "values most" means "would pay most", and what someone can pay depends on their income. A tourist outbidding a young local for the last flat in Gràcia is efficient by this measure and may still strike you as unfair; economics measures the cake, you judge the slices.

A tax of 0.90 a cup: buyers pay 2.50, the bars keep 1.60



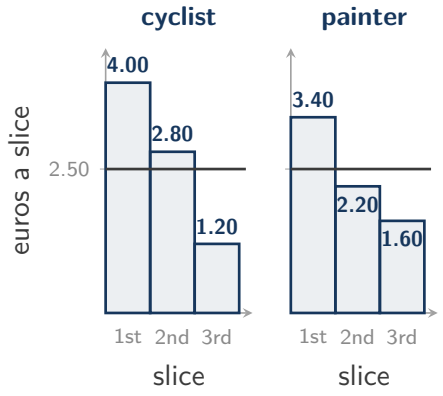
Buyers pay 0.50 more, the bars keep 0.40 less: a tax is a wedge between two prices.

Slide 13 A tax of 0.90 a cup: buyers pay 2.50, the bars keep 1.60

A tax on every cup sold, handed in by the bars, is an indirect tax: a tax on spending rather than on income. Supply moves up by exactly the tax, because a bar now needs 0.90 more from the customer before any given cup is worth making; it is Class 6's supply shift, drawn again. At the new crossing the buyers' price is read on the shifted line and the bars' price, what they keep after handing over the tax, on the old line directly below it: the tax opens a wedge between two prices, and the market shrinks to the quantity where that wedge just fits between the curves, slide 10's gap.

Neither side pays the whole 0.90: the price rises by less than the tax, and the bars bear part of it although they are the ones who hand it in. Why buyers carry a little more than half is Class 7's elasticities, made into a rule after the break.

Market demand from two staircases (optional)



Pizza slices: two buyers

1. At 2.50: slices bought in all?
2. Their consumer surplus at 2.50?
3. Market demand at 2.00?

Slide 14 Market demand from two staircases (optional)

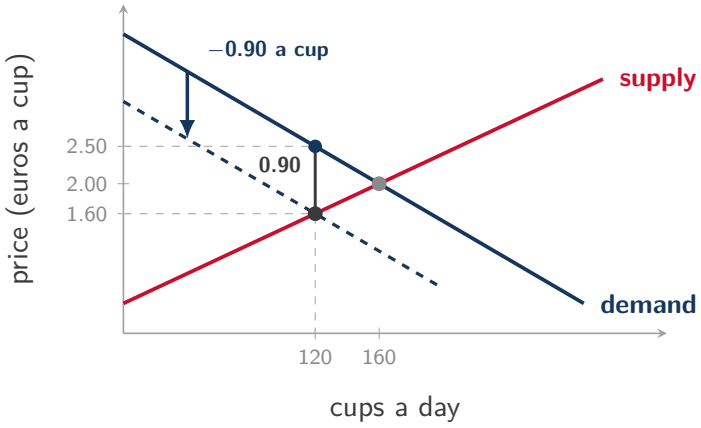
An optional exercise on building market demand from two buyers' staircases, as on slide 5. Before class, read off each graph how many slices each buyer takes at the price line, then combine the two buyers at the same price; for the surplus, take each slice bought and compare what it is worth with the price. Then do the same at the lower price of item 3.

Revealed in class.

Break

Ten minutes

Tax the buyers instead: the same 120 cups, the same 2.50 and 1.60



Who hands the 0.90 to the tax office does not matter: the wedge does.

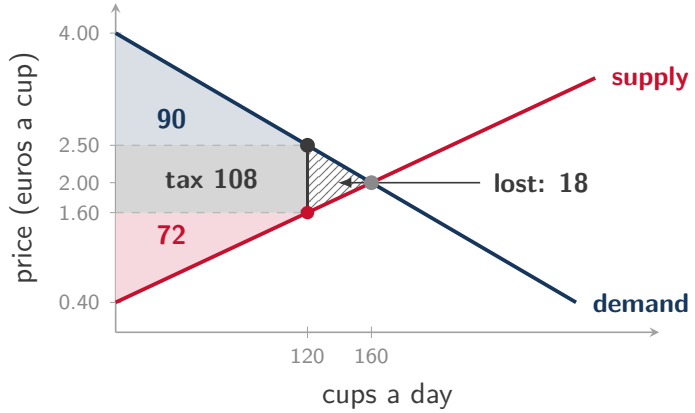
Slide 15 Break

Ten minutes. After it: who hands the tax to the tax office, and where the money goes.

Slide 16 Tax the buyers instead: the same 120 cups, the same 2.50 and 1.60

The same tax, collected from the other side. If customers pay the 0.90 themselves, a cup with a counter price P costs them P plus 0.90, so what each buyer is willing to pay at the counter falls by exactly the tax: demand moves down by 0.90 instead of supply moving up. The new crossing gives the counter price, which is what the bars keep; adding the tax back gives what buyers pay in all, and the three numbers are slide 13's. The law chooses who hands the money over; it cannot choose who bears it, because the wedge between the two prices is the same whichever curve moves. Most governments collect from sellers simply because there are fewer of them to check.

With the tax: buyers 90, bars 72, the state 108 — and 18 lost



Buyers lose 70, the bars 56; the state gets 108. The lost 18: the **deadweight loss**, Thursday.

Slide 17 With the tax: buyers 90, bars 72, the state 108, and 18 lost

Slide 9's 288, after the tax. The buyers' triangle now starts at the higher price and the bars' triangle ends at the lower one, so both shrink; the rectangle between the two prices, as tall as the tax and as wide as the cups still sold, is what the state collects. The three areas add up to 270, not 288: the missing 18 went to nobody, because the forty cups between 120 and 160 are no longer made, and slide 10 showed that each of them was worth more than it cost. That triangle is the deadweight loss, the fall in total surplus that a tax causes; today it gets its name and its picture, and Thursday's reading is about it.

The trap is to count the state's 108 as a loss: that money pays for something, and the loss to everyone together is only the 18.

In pairs, four minutes: a tax of 2.50 on every menú

Same street, same menús

menús a day	the last one is worth	the last one costs
30	15.00	10.00
45	13.50	11.00
60	12.00	12.00

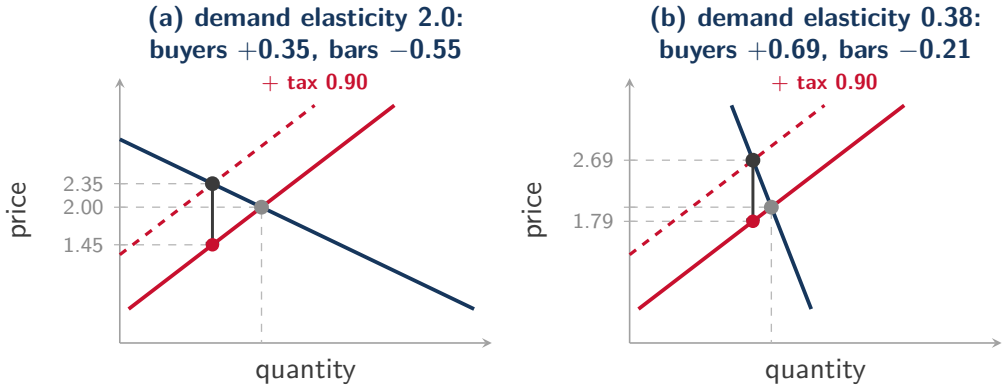
1. No tax: price and menús?
2. Tax 2.50: menús, and each side's price?
3. Tax collected a day?
4. Who bears more of the 2.50?

Slide 18 In pairs, four minutes: a tax of 2.50 on every menú

Exercise 2 is slide 8's street as a table: "worth" is the height of demand at that quantity and "costs" the height of supply. Before class, find the row where the market stops without a tax, then the row where the gap between worth and cost equals the tax, and read each side's price from it. Count the tax on the menús still sold, and compare each side's price with the no-tax price.

Revealed in class.

Class 7's markets, taxed 0.90: who cannot walk away pays more



Buyers who can switch leave most of the tax to the bars; buyers who cannot pay most of it.

Slide 19 Class 7's markets, taxed 0.90: who cannot walk away pays more

Class 7's last figure, read again. The dashed line is the same supply shift as then, and today it has a cause: a tax of 0.90 a cup, with the supply curve identical in both panels. In (a) buyers can switch, so when the tax arrives they leave: the price they pay rises only a little and the bars absorb most of the 0.90. In (b) buyers have few substitutes, so most stay and pay: the price rises by most of the tax and the bars keep almost what they kept before. Nothing on the sellers' side differs between the panels, so the whole difference in who bears the tax comes from Class 7's elasticity of demand.

Tax incidence: the less elastic side bears more of the tax

	most of the tax falls on
demand less elastic	buyers
supply less elastic	sellers
equally elastic	half each

The law says who pays the tax office; elasticities say who bears the tax.

Slide 20 Tax incidence: the less elastic side bears more of the tax

The word for how the burden of a tax is shared is tax incidence, and the table is the rule the last two slides illustrated. An elasticity measures how easily a side can leave the market when the deal gets worse; the side with fewer alternatives stays, and whoever stays bears the tax. The coffee tax, the menú tax and the two panels all fit the rule once the two elasticities are known. Who is named in the law is a separate question altogether, as slide 16 showed.

In pairs, three minutes: who bears most of the tax?

Buyers or sellers — and why?

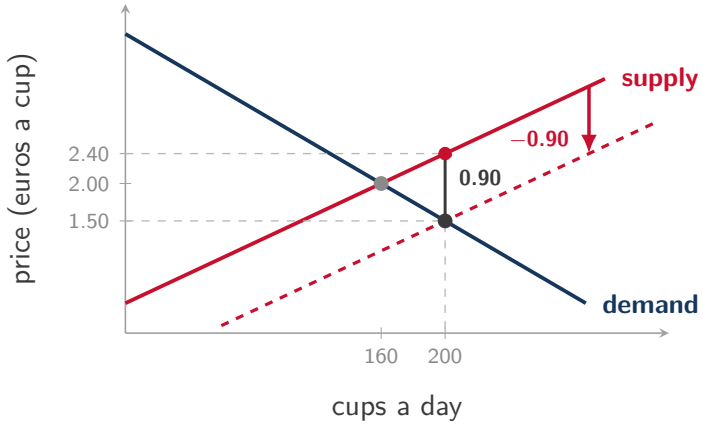
1. Cigarettes: demand elasticity 0.4, supply 2.0
2. Beach umbrellas on a July Sunday: demand 3.0, supply 0.5
3. Tax 1.00; buyers' price 5.00 → 5.80. Sellers keep? Less elastic side?
4. The same tax, paid by the buyers: price on the label?

Slide 21 In pairs, three minutes: who bears most of the tax?

Exercise 3 applies the incidence rule four times. Before class, decide for items 1 and 2 which side has the smaller elasticity and therefore stays in the market. For item 3, work out what sellers keep from the tax and the buyers' price, and compare each side's move with the old price. For item 4, think back to slide 16 and ask what the price on the label would show if the buyers handed the tax over.

Revealed in class.

A subsidy of 0.90 a cup: buyers pay 1.50, the bars get 2.40



The tax reversed: buyers save 0.50, the bars gain 0.40.
The state pays $0.90 \times 200 = 180$ € a day.

Slide 22 A subsidy of 0.90 a cup: buyers pay 1.50, the bars get 2.40

A subsidy is a tax with the sign changed: the state pays the bars a fixed sum for every cup, so a bar needs that much less from the customer and supply moves down by the subsidy. The market grows past 160, buyers pay less and the bars receive more, and the gap between the two prices is again 0.90, now paid by the state rather than collected by it. The split is the same as under the tax, because the same two elasticities decide it: a subsidy paid to sellers does not stay with the sellers.

The bill is the subsidy times every cup sold, including the cups that were sold anyway; and the extra cups past 160 cost more than they are worth, slide 10's 200th cup, so a subsidy has a lost triangle of its own. Class 5's T-usual fare is a subsidy of this kind.

Per unit or per cent: VAT is a percentage of the price

- A fixed sum per unit: a **specific tax** (0.90 a cup)
- A share of the price: **ad valorem** (VAT)
- Spain's VAT: 21 %, 10 %, 4 %
- A loaf: 1.00 + 4 % VAT = 1.04 €

Either way a wedge; the elasticities decide who bears it.

Slide 23 Per unit or per cent: VAT is a percentage of the price

Two ways to write a tax. A specific tax is a fixed sum per unit, like today's 0.90 a cup or the tax on a litre of fuel; an ad valorem tax is a share of the price, so its wedge is wider where the price is higher, which is why the reading draws supply turning rather than shifting. Spain's three rates put most goods at the top one, restaurants and some foods in the middle, and the basics at the bottom, where slide 4's loaf sat. The rule of slide 20 does not care which kind of tax it is: either way there is a wedge, and the elasticities decide who bears it. The next slide takes that rule back to the bread.

Bread, milk, eggs: what reached the shoppers

Revealed in class

Before class: by slide 20's rule, who should get most of a VAT cut on bread — shoppers or shops? Bread is a necessity, and a supermarket can stock more at about the same cost.

Slide 24 Bread, milk, eggs: what reached the shoppers

Revealed in class.

Beforehand: take slide 4's question to slide 20's rule. Bread is a necessity, so ask how elastic its demand is; a supermarket can stock more at about the same cost, so ask how elastic its supply is. Then say which side should have borne the VAT, which side should get most of a cut, and why.

Which curve moves? (optional)

Cinema tickets in one city

1. A tax of 1.00 a ticket, paid by the cinemas: which curve shifts?
2. Tickets get dearer: does demand shift too?
3. No tax; a subsidy makes streaming cheaper. Ticket demand?
4. And the ticket price, and the tickets sold?

Slide 25 Which curve moves? (optional)

An optional exercise on shifts and moves along a curve, in the market for cinema tickets. Before class, decide for each event which curve it shifts, if any, and which way; then ask what the change in the ticket price does on the other curve. Item 3 is about another good's price: think back to substitutes in Class 5.

Revealed in class.

Who gains from trade? (optional)

Hours for one unit

	1 kg of cheese	1 bottle of wine
valley farm	2	1
hill farm	3	6

1. Who should make the cheese, and why?
2. At 1 bottle a kg: each farm's gain?
3. If the hill farm took 4 h a kg, 2 h a bottle?

Slide 26 Who gains from trade? (optional)

An optional exercise linking comparative advantage (Block 1) with today's surplus. Before class, work out for each farm what a kilo of cheese costs in bottles of wine given up, and decide who should make the cheese, even though one farm is faster at both. Then ask how a price between the two costs shares the gain, and what changes in item 3.

Revealed in class.

Surplus: value minus price for buyers, price minus cost for sellers.

The market's quantity makes total surplus as large as it can be.

A tax or a subsidy is a wedge; the less elastic side feels most of it.
Deadweight loss and price controls: Thursday.

Slide 27 In one line

The class in four lines, to read slowly: line two is slide 11's answer to slide 3, line three is the rule of slide 20, and line four names Thursday.

Before Thursday 22 October

- Thursday: price controls; international trade and tariffs
- Read Mankiw & Taylor: Ch. 7, pp. 159–163 and 172–178; Ch. 17, pp. 380–390
- Problem set 1: due Thursday 22 October, on paper, in class
- Practise in Sextant: the units the course page lists

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Slide 28 Before Thursday 22 October

Thursday 22 October: what happens when the government sets the price itself, rent caps and price floors, then the deadweight loss properly, and international trade and tariffs. Reading: Mankiw & Taylor chapter 7, pages 159 to 163 and 172 to 178, and chapter 17, pages 380 to 390. Problem set 1, handed out on 15 October, is due on Thursday 22 October, on paper, in class, one per person. Practise in Sextant with the units the course page lists.

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